Appendix

RETURNS

1. Returns

There are several definitions of returns. This appendix shows how they are related to each other.

1.1. GROSS RETURNS

A *gross return* over one period, t to t+1, is defined as

$$\text{Gross return } R_{t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}, \tag{A.1}$$

where P_t is the price of the asset at the beginning of the period, P_{t+1} is the price at the end of the period, and D_{t+1} is any dividend paid at time t+1. I denote gross returns with an upper case R. The definition in equation (A.1) assumes that we can measure prices—which is not true for some illiquid investments. For now, we assume assets are traded; I discuss illiquid asset returns in chapter 13.

As wealth cumulates over time, we compound gross returns multiplicatively. An asset earning gross returns of $R_{t+1} = 1.10$ three years in a row produces cumulated wealth at the end of three years of

$$\begin{aligned} W_{t+3} &= R_{t+1}R_{t+2}R_{t+3} \\ &= 1.10 \times 1.10 \times 1.10 \\ &= 1.3310, \end{aligned}$$

starting with \$1 at time t.